

The Bosonic Pt Particle: in M^D

$$S = -m \int dt, \quad ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = -d\tau^2$$

$$\therefore dt^2 = -\eta_{\mu\nu} dx^\mu dx^\nu \Rightarrow S = -m \int \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

$$S = -m \int d\tau \sqrt{-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu}$$

- Point Particle extremizes its proper time
- ~~Parametrization~~: $\tau \rightarrow \tau'(\sigma) = \tau - f(\tau)$

$$x^\mu(\tau') = x^\mu(\tau) \quad (\text{Scalar})$$

$$\text{Variation: } \delta\phi(\xi) = \phi'(\xi) - \phi(\xi)$$

$$\delta x^\mu = x^\mu(\tau) - x^\mu(\tau')$$

$$x'^\mu(\tau) = x^\mu(\tau') + f(\tau') \frac{d}{d\tau'} x^\mu(\tau')$$

$$\therefore \delta x^\mu(\tau) = f(\tau) \dot{x}^\mu(\tau)$$

Feynman formulation: $S = \frac{1}{2} \int d\tau \{ e^{-1} \dot{x}^\mu \dot{x}^\nu \eta_{\mu\nu} - m^2 e \}$

x^μ & e are separate fields.

$$\Rightarrow e_\mu^a = \frac{\partial x^a}{\partial \tau^b} e_\mu^b$$

$$x^a = \tau^a$$

$$\boxed{\delta e = f \dot{e} + \dot{f} e}$$

$$e_\mu^{a'}(\tau') = (1 - \dot{f}) e_\mu^b$$

$$e_\mu^{a'}(\tau) = f \dot{e}_\mu^a = (1 - \dot{f}) e_\mu^b$$

$$\delta e_\mu^a = e_\mu^{a'}(\tau) - e_\mu^a(\tau) = f \dot{e} + \dot{f} e$$

* Scalar field Coupled to gravity

$$\int d^D x \, e \, \partial_\mu x^a \partial^\mu x_a, \quad \text{in 1D } \mu \rightarrow t$$

$$\int d^D x \, e \, \partial_t x^a \partial_t x_a$$

and since this is 1-D. we have

$$e_\mu^t \eta_{tt} e_\nu^t = g_{\mu\nu} \quad \text{and so the } \delta(\det(e_\mu^a)) = e_\mu^a$$

and thus $g_{\mu\nu}$ must be 1D as well

$$e^\tau \eta_{\tau\tau} e^\tau = g_{\tau\tau} \quad e^\tau \equiv e$$

$$-e^2 = g_{\tau\tau}$$

- Now express the action to $\mathcal{O}(\dot{x}^m)$ and introduce conj. momenta $P_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$

$$P_\mu = e^{-1} \dot{x}^\nu \eta_{\mu\nu}$$

we can solve for e using the on-shell

$$\text{condition } P_\mu P^\mu = -m^2$$

$$(e^{-1} \dot{x}^\nu \eta_{\mu\nu}) (e^{-1} \eta^{\mu\alpha} \dot{x}^\alpha \eta_{\alpha\sigma}) = -m^2$$

$$e^{-2} \dot{x}^\nu \eta_{\mu\nu} \dot{x}^\mu = -m^2$$

- Actually not sure... but we have

$$S = \int d\tau \left\{ \dot{x}^\mu P_\mu \eta_{\mu\nu} - \frac{e}{2} (P^\mu P_\mu \eta_{\mu\nu} + m^2) \right\}$$

Recover previous form by using P_μ E.O.M.

i.e. the previous is on-shell.

$$\text{E.O.M: } \frac{1}{2} \int d\tau (\dot{e} \eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu - m^2 e)$$

$$\frac{\partial}{\partial e} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = 0$$

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial e} \right) - \frac{\partial \mathcal{L}}{\partial e} = 0$$

$$\frac{d}{d\tau} \left(-\frac{1}{2} \frac{1}{e^2} \dot{x}^2 - m^2 \right) = 0 \Rightarrow \frac{d}{d\tau} \left(\frac{\dot{x}}{e} \right) = 0$$

$$\text{or } \frac{d}{d\tau} (e^{-1} \dot{x}^\mu) = 0$$

$$e^2 m^2 + \dot{x}^2 = 0$$

and the mass-shell condition. $P^2 = -m^2$

⊗: substituting this into the above we find

$$e = \frac{1}{m} \sqrt{-\dot{x}^2}$$

$$S = \frac{1}{2} \int d\tau \left(\frac{m}{\sqrt{-\dot{x}^2}} \dot{x}^2 - m^2 \left(\frac{\sqrt{-\dot{x}^2}}{m} \right) \right)$$

$$= \frac{1}{2} \int d\tau (-m \sqrt{-\dot{x}^2} - m \sqrt{-\dot{x}^2}) = -m \int d\tau \sqrt{-\dot{x}^\mu \dot{x}^\mu \eta_{\mu\nu}}$$

Which is the original action.

⊗ Dir: Poincaré invariant,

$S_{2/3}$: Can be used for massless particles

$$m=0.$$

• Also invariant under CTT's

For $m=0$,

* From S_1 we have

$$p_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = \frac{m \dot{x}_\mu}{\sqrt{-\dot{x}^\mu \dot{x}_\mu}} \eta_{\mu\nu}$$

and so $p^2 + m^2 \equiv \phi = 0$

Note: $\dot{x}^\mu \dot{x}_\mu = \dot{x}^i \dot{x}_i$

$\vec{v} = (\dot{x}_1, \dot{x}_2, \dot{x}_3) \Rightarrow |\vec{v}|^2 = \dot{x}^i \dot{x}_i$

$(\dot{x}^0)^2 = \left(\frac{dt}{d\tau}\right)^2$ Affine time, then $=1$

and we are left with

$p_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = m \dot{x}_\mu / \sqrt{1 - v^2} = \gamma m \dot{x}_\mu \quad (c=1)$

BRST Quantization Review: $[T_a, T_b] = f_{ab}^c T_c$

$\Delta_\mu = g A_\mu^a T_a \mid \text{Tr}(T_a T_b) = -C_F \delta_{ab}$

$F_{\mu\nu} = g F_{\mu\nu}^a T_a = [D_\mu, D_\nu]$

$S_{YM} = \frac{1}{2g^2 C_F} \int d^4x \text{Tr}(F_{\mu\nu} F^{\mu\nu}) = -\frac{1}{2} \int d^4x F_{\mu\nu}^a F_a^{\mu\nu}$

• Gauge fixing condition $G(A_\mu) = 0$

(i) $\partial_\mu A^\mu = 0 = G(A_\mu)$ (Gauge choice)

→ We incorporate this into the action since

$G(A_\mu) = 0$ then, of course

$S_{YM} + \int \mathcal{L}(G(A_\mu)) = S_{YM}$

We do so by introducing a (Bosonic) Lag. multiplier.

$S^{gf} = - \int d^4x \text{Tr}(B G(A_\mu)) \mid B = B^a T_a \in \mathfrak{g} \otimes \mathbb{C}$

* The new action is given by

$S = S_{YM} + S^{gf}$

this action is not Gauge invariant under

$A_\mu \rightarrow U^\dagger A_\mu U + U^\dagger \partial_\mu U \mid U = \exp(i\omega^a(x) T_a)$
 $\left\{ \begin{aligned} & e^{-i\omega T} A_\mu e^{i\omega T} + e^{-i\omega T} \partial_\mu (e^{i\omega T}) \end{aligned} \right.$

$$\begin{aligned}
 &= (1 - \omega T_a) A_m^b T_b (1 + \omega^c T_c) + \\
 &= (A_m^b T_b - \omega^a A_m^b T_a T_b) + A_m^b T_b \omega^c T_c \\
 &\quad + (1 - \omega^d T_d) \delta_m \omega^e T_e (1 + \omega^f T_f) \\
 &= A_m + \delta_m \omega \leftarrow \text{"Usual" Gauge trans.}
 \end{aligned}$$

Gauge invariance will be replaced by the BRST symmetry.

* Take $U = \exp\{\omega^a T_a\} \mapsto \exp\{\Lambda^a T_a\}$
i.e. $\omega^a \rightarrow \Lambda^a$.

where c^a : Grassmann odd parameter (Ghost)

$$\delta A_m = D_m \omega \mapsto \delta A_m = \Lambda D_m c$$

$$\delta A_m = \Lambda D_m c \equiv \Lambda (D_m c + [A, c])$$

Ghost transformations:

$$\delta c = -\Lambda c \cdot c = -\frac{\Lambda}{2} \{c, c\} \quad \left| \quad c \cdot c = \begin{pmatrix} \text{matrix} \\ \text{mult.} \end{pmatrix} \right.$$

$$\delta c^e = -\frac{\Lambda}{2} c^a c^b f_{ab}^e$$

Anti-Ghost transf: $b \equiv b^a T_a$

$$\delta b = \Lambda b, \quad \delta B = 0$$

$$\delta_{\text{BRST}}^2 = 0$$

* The full action:

$$S_{\text{gh}} = -\int d^4x \text{Tr} \{ b(x) \delta G(x) \}$$

$$\text{But } \delta G(x) = \int d^4y \frac{\delta G(x)}{\delta A^a(y)} \delta A^a(y)$$

$$\text{and } \delta A^a(x) = \Lambda D^a c$$

Thus:

$$S_{\text{gh}} = -\int d^4x \text{Tr} \left\{ b(x) \int d^4y \frac{\delta G(x)}{\delta A^a(y)} \Lambda D^a c \right\}$$

The full Action reads

$$S = S_{\text{YM}} + S_{\text{gh}} + S_{\text{gf}}$$

$$S = \frac{1}{2g^2 C_F} \int d^4x \text{Tr} \{ F_{\mu\nu} F^{\mu\nu} \} - \int d^4x \text{Tr} \{ B(x) G(A_m) \}$$

$$= \int d^4x \text{Tr} \{ b(x) \int d^4y \frac{\partial \Lambda}{\partial x^\mu} \Lambda D^\mu c(y) \}$$

* Lorentz Gauge we have: $G = \partial^\mu A_\mu$

$$\begin{aligned} S^{\text{gh}} &= - \int d^4x \text{Tr} \{ b(x) \int d^4y \partial_\mu^\alpha \delta(x-y) \Lambda D^\mu c(y) \} \\ &= + \int d^4x \text{Tr} \{ b(x) \int d^4y \delta(x-y) \partial_\mu^\alpha (\Lambda D^\mu c(y)) \} \\ &= \int d^4x \text{Tr} \{ b(x) \Lambda \partial_\mu D^\mu c(x) \} \end{aligned}$$

* Note: $\Delta(S^{\text{gh}} + S^{\text{gf}}) = - \int d^4x \delta_\Lambda \text{Tr} \{ b(x) G[A^\mu(x)] \}$

$f(\tau) \rightarrow \Delta c(\tau)$, Δ grassmann odd

(Recall $\tau' = \tau - f(\tau)$) and

$$\delta x^\mu = f \dot{x}^\mu, \quad \delta c = f \dot{c} + \dot{f} c$$

$$\delta x^\mu = (\Delta c) \dot{x}^\mu, \quad \delta c = \frac{d}{d\tau} (\Delta c)$$

* For Δc to be real i.e. $(\Delta c)^\dagger = \Delta c$

we take $c^\dagger = c$, $\Delta^\dagger = -\Delta$

$$(\Delta c)^\dagger = c^\dagger \Delta^\dagger = -c \Delta = \Delta c \quad \checkmark$$

$$\delta x^\mu = (\Delta c) \dot{x}^\mu \quad \text{and} \quad \delta^2 x^\mu = 0$$

$$0 = \Delta \delta c \dot{x}^\mu + \Delta \dot{c} \delta x^\mu$$

$$\Delta \delta c \dot{x}^\mu + \Delta c (\Delta \dot{c} \dot{x}^\mu) + (\Delta c) (\Delta c) \dot{x}^\mu = 0$$

$$\Delta^2 c \dot{x}^\mu + c \Delta \dot{c} \dot{x}^\mu + c \Delta c \dot{x}^\mu = 0$$

$$\delta c \dot{x}^\mu + c \Delta \dot{c} \dot{x}^\mu + c^2 \Delta \dot{x}^\mu = 0$$

but $c^2 = 0$ so

$$\delta c \dot{x}^\mu = -c \Delta \dot{c} \dot{x}^\mu$$

$$\delta c = -c \Delta \dot{c} \quad \text{or}$$

$$\boxed{\delta c = (\Delta c) \dot{c}}$$

$$[\delta_1, \delta_2] x^\mu = \delta_1(f_2 \dot{x}^\mu) - \delta_2(f_1 \dot{x}^\mu)$$

$$\begin{aligned}
&= f_2 (f_2 \dot{x}^\mu + f_1 \ddot{x}^\mu) \\
&\quad - f_1 (f_2 \dot{x}^\mu + f_2 \ddot{x}^\mu) \\
&= f_2 f_1 \dot{x}^\mu + f_2 f_2 \ddot{x}^\mu - f_1 f_2 \ddot{x}^\mu \\
&\quad - f_1 f_2 \ddot{x}^\mu \\
&= f_{12} \ddot{x}^\mu
\end{aligned}$$

$$\therefore f_{12} \equiv f_2 \dot{f}_1 - f_1 \dot{f}_2$$

Since $\delta C^a = \frac{1}{2} \tilde{f}_2^a$, $[\delta_1, \delta_2] = \delta_{f_{12}}$

Taking $f \rightarrow \Lambda C$

$$\delta \lambda = 0, \delta b = \Lambda \lambda$$

*Gauge fixing terms:

$$\therefore e(\tau) = 1$$

$$\begin{cases} G = \ln(e) = 0 \\ \text{or } G(\tau) = 1 - e \end{cases}$$

$$\bullet \Delta^{\text{gf}} = \int d\tau \lambda \ln(e)$$

$$S^{\text{BRST}} = \frac{1}{2} \int d\tau \{ e^{-2} \dot{x}^\mu \dot{x}^\nu \eta_{\mu\nu} - m^2 e \} + \int d\tau \lambda \ln(e) - \int d\tau b D_\tau C$$

$$D_\tau C \equiv \partial_\tau C + \frac{d(\ln(e))}{d\tau} C$$

Quantum theory: $\int D e D x^\mu D C D b D \lambda e^{i \Delta^{\text{BRST}}}$

$$\begin{aligned}
&\underline{\lambda = 1}: \int D e \dots \left(\int D \lambda e^{\int d\tau \lambda \ln(e)} \right) e^{i(S^{\text{orb}} + S^{\text{gh}})} \\
&= \int D e \dots \delta(\ln(e)) e^{i(S^{\text{orb}} + S^{\text{gh}})}
\end{aligned}$$

This sets $\ln(e) = 0 \Rightarrow e = 1$ ✓

$$\int D x^\mu D C D b \exp \left\{ i \left(\frac{1}{2} \int d\tau \{ \dot{x}^\mu \dot{x}^\nu \eta_{\mu\nu} - m^2 \} - \int d\tau b D_\tau C \right) \right\}$$

Further:

$$D_\tau e = \dot{e} + \frac{1}{e} \frac{de}{d\tau} C, \quad \frac{de}{d\tau} = 0$$

$$D_\tau C = \dot{C}$$

$$Z[b, C] = \int D x^\mu D C D b e^{i \left(\frac{1}{2} \int d\tau \{ \dot{x}^\mu \dot{x}^\nu \eta_{\mu\nu} - m^2 \} - \int d\tau b \dot{C} \right)}$$

(We must replace λ in: $\delta b = \Lambda \lambda$)

Use e E.O.M

$$\partial_\tau \frac{\partial \mathcal{L}}{\partial \dot{c}} - \frac{\partial \mathcal{L}}{\partial c} = 0$$

$$\delta \mathcal{L} / \delta c = -\frac{1}{2} \dot{c}^2 \dot{x}^\mu \dot{x}_\mu - \frac{1}{2} m^2 + \lambda \frac{1}{c}$$

$$\partial \mathcal{L} / \partial \dot{c} = b e^{-1} \dot{c} \quad | \quad e=1$$

$$\therefore \partial_\tau () = \frac{d}{d\tau} (b \dot{c})$$

$$\therefore -\frac{d}{d\tau} (b \dot{c}) - \left(-\frac{1}{2} \dot{x}^\mu \dot{x}_\mu - \frac{1}{2} m^2 + \lambda \right) = 0$$

$$\therefore \lambda = \frac{d}{d\tau} (b \dot{c}) + \frac{1}{2} \dot{x}^\mu \dot{x}_\mu + \frac{1}{2} m^2$$

$\therefore$

$$\boxed{\delta b = \Lambda \lambda = \Lambda \left(\frac{1}{2} \dot{x}^\mu \dot{x}_\mu + \frac{1}{2} m^2 - \frac{d}{d\tau} (b \dot{c}) \right)}$$

Noether charge:

$$\boxed{S^{\text{total}} = \int d\tau \left(\frac{1}{2} \dot{x}^\mu \dot{x}_\mu - \frac{1}{2} m^2 - b \dot{c} \right)}$$

with

$$\delta x^\mu = (\Lambda c) \dot{x}^\mu$$

$$\delta b = \Lambda \left(\frac{1}{2} \dot{x}^\mu \dot{x}_\mu + \frac{m^2}{2} - \frac{d}{d\tau} (b \dot{c}) \right)$$

$$\delta c = (\Lambda c) \dot{c}$$

* Use Λ as the symm. parameter:

$$\delta_{\text{total}} S^{\text{total}} = \int d\tau \left\{ \dot{x}^\mu (\delta x_\mu) - \delta b \dot{c} - b (\delta \dot{c}) \right\}$$

$$= \int d\tau \dot{x}^\mu \Lambda (\dot{c} \dot{x}_\mu + c \ddot{x}_\mu) = \Lambda (\dot{c}) \dot{c} - \dots$$

$$= \int d\tau \left(\dot{x}^\mu \Lambda \dot{c} \dot{x}_\mu + \dot{x}^\mu \Lambda c \ddot{x}_\mu - \Lambda \left(\frac{1}{2} \dot{x}^\mu \dot{x}_\mu \dot{c} + \frac{m^2}{2} \dot{c} - \frac{d}{d\tau} (b \dot{c}) \right) - b \Lambda \frac{d}{d\tau} (c \dot{c}) \right)$$

$$= \int d\tau - \Lambda c (\dot{x}^\mu \dot{x}_\mu) - \dot{x}^\mu \Lambda c \ddot{x}_\mu + \frac{1}{2} \Lambda \dot{x}^\mu \dot{x}_\mu + \underline{\Lambda b c} + b \underline{\Lambda c \dot{c}}$$

$\Rightarrow$

$$= \Lambda \left(2 \dot{x}^\mu \dot{x}_\mu c + \frac{1}{2} \dot{x}^\mu \dot{x}_\mu c + m^2 c \right)$$

* These factors don't match, the $2 \rightarrow \frac{1}{2}$
 ... Not sure how...
 $p^\mu \equiv \dot{x}^\mu$

$$\therefore J^T = c(p^\mu p_\mu + m^2)$$

There are no SPATIAL Dim to int. over thus

$$Q = \int J^T = \delta^T \text{ is}$$

$$Q = c(p^\mu p_\mu + m^2)$$

Poisson Bracket for Spm $[-1/2, 1/2]$:

Configuration Space (q_A, p_A)

$$\{f, g\}_{P.B.} \equiv \sum_A \left\{ f \frac{\overrightarrow{\partial}}{\partial q_A} \frac{\overrightarrow{\partial}}{\partial p_A} g - (-1)^{fg} g \frac{\overrightarrow{\partial}}{\partial q_A} \frac{\overrightarrow{\partial}}{\partial p_A} f \right\}$$

$$(-1)^{fg} \equiv \begin{cases} -1 & \text{if } f \text{ \& } g \text{ are Grassmann odd} \\ +1 & \text{otherwise} \end{cases}$$

$$\{f, g\}_{P.B.} = -(-1)^{fg} \{g, f\}_{P.B.} \quad | \quad \{f, g\}_{P.B.} = (-1)^{fg} g \{f, a\}_{P.B.} + \{f, g\}_{P.B.}$$

$$\{f_1 + f_2, g\}_{P.B.} = \{f_1, g\}_{P.B.} + \{f_2, g\}_{P.B.}$$

Point Particle: $\{x^\mu, p^\mu\}_{P.B.} = \eta^{\mu\nu} \quad | \quad \{c, b\}_{P.B.} = 1$

$$H = p^\mu p_\mu + m^2$$

$$\delta(\cdot) = \{ \cdot, H \}_{P.B.} \quad | \quad \{Q, Q\} = 0$$

Quantization: (Dirac Rule)

$$\{ \cdot, \cdot \}_{P.B.} \mapsto \begin{cases} \frac{1}{i\hbar} \{ \cdot, \cdot \} & \text{for two odd Quantities} \\ \frac{1}{i\hbar} [\cdot, \cdot] & \text{otherwise} \end{cases}$$

$$\therefore [x^\mu, \tilde{p}^\nu] = i\hbar \eta^{\mu\nu} \quad | \quad \{c, b\} = i\hbar$$

In the Schrödinger rep:

$$x^\mu \rightarrow x^\mu, \quad c \rightarrow c, \quad p^\mu \rightarrow -i\hbar \frac{\partial}{\partial x^\mu}, \quad b \rightarrow i\hbar \frac{\partial}{\partial c}$$

$$* Q = c(p^\mu p_\mu + m^2) \rightarrow \tilde{Q}$$

$$\tilde{Q} = \tilde{c}(-\partial^2 + m^2)$$

NilPotency comes from the fact that $\tilde{c}^2 = 0$.

$\Psi(x^\mu, c) = \text{wave f}^u$

- Expand in c

$$\bar{\Psi}(x^\mu, c) = \bar{\Psi}(x^\mu) + c \frac{\partial \bar{\Psi}(x^\mu, c)}{\partial c}$$

Physical states: satisfy the BRST cohomology

i.e. we get rid of redundant counting

$$\psi_1 \sim \psi_2 \Leftrightarrow \exists \psi_3 \text{ s.t.}$$

$$\psi_1 - \psi_2 = Q \psi_3$$

- Note, for a state to be invariant under a BRST trans.

$$e^{iQ} |\psi\rangle = |\psi\rangle$$

$$\text{or } e^{iQ} |\psi\rangle = |\psi\rangle + iQ |\psi\rangle$$

Thus, the physical Requirement is that

$$Q |\psi\rangle = 0$$

But $Q = c(-\partial^2 + m^2)$, so in field theory this is equivalent to

$$c(-\partial^2 + m^2) \psi = 0, \quad c \neq 0$$

$$[-\partial^2 + m^2] \psi = 0$$

The Super Point Particle:

(i) Spinning Particle: World-line susy

(ii) Brink-Schwarz Particle: Space time susy

- Begin with a theory with D fields x^μ and their susy partners χ^μ

Consider: $S^F = \frac{1}{2} \int d\tau (\dot{x}^\mu \dot{x}^\mu - i \dot{\chi}^\mu \chi^\mu) m_{\text{world}}$

(i) Time translations: $\tau = \tau' + a$

$$\dot{x}^{\mu'}(\tau) = \dot{x}^{\mu}(\tau^0) + a \dot{\tilde{x}}^{\mu}(\tau^0)$$

$$\delta x^{\mu} = a \dot{\tilde{x}}^{\mu}(\tau)$$

$$\delta \dot{x}^{\mu} = a \ddot{\tilde{x}}^{\mu}(\tau)$$

$$\{Q_{\alpha}, Q_{\beta}^{\dagger}\} = i\sigma_{\alpha\beta}^{\mu} P_{\mu}$$

Rigid susy: $\delta x^{\mu} = i\epsilon \dot{x}^{\mu}$, $\delta \dot{x}^{\mu} = \dot{x}^{\mu} \epsilon$

$$\epsilon^* = \epsilon, \quad \chi^* = \chi$$

We note

$$[\delta_1, \delta_2](\cdot) = 2i\epsilon_1 \epsilon_2 \frac{d}{d\tau}(\cdot)$$

Commutator of susy $\sim$ time translation
of magnitude $2i\epsilon_1 \epsilon_2$

$$\delta x^{\mu} = i\epsilon \dot{x}^{\mu} \Rightarrow \delta \dot{x}^{\mu} = i\dot{\epsilon} \dot{x}^{\mu} + i\epsilon \ddot{x}^{\mu}$$

$$\delta \dot{x}^{\mu} = \dot{x}^{\mu} \epsilon \Rightarrow \delta \ddot{x}^{\mu} = \ddot{x}^{\mu} \epsilon + \dot{x}^{\mu} \dot{\epsilon}$$

$$\delta S^F = \int d\tau \left(i\epsilon \dot{x}^{\mu} \dot{\tilde{x}}^{\nu} \eta_{\mu\nu} - i\dot{\tilde{x}}^{\mu} \epsilon \dot{x}^{\nu} \eta_{\mu\nu} \frac{1}{2} \right. \\ \left. + \frac{1}{2} i\dot{x}^{\mu} \dot{\tilde{x}}^{\mu} \epsilon + \frac{1}{2} i\dot{x}^{\mu} \dot{\tilde{x}}^{\mu} \dot{\epsilon} \right)$$

$$\int \delta S^F = \dot{\tilde{x}}^{\mu} \dot{x}^{\nu} \eta_{\mu\nu}$$

* If we now couple the theory to the worldline
Supergravity $(e, \psi) =$

$$S = \frac{1}{2} \int d\tau (e^{-1} \dot{x}^{\mu} \dot{x}^{\nu} - i \dot{x}^{\mu} \dot{\tilde{x}}^{\nu} - R e^{-1} i \psi \dot{x}^{\mu} \dot{\tilde{x}}^{\nu}) \eta_{\mu\nu}$$

* This means we must introduce local susy
terms:

$\delta x^m = i\epsilon \dot{x}^m$	$\delta x^m = (\dot{x}^m - \frac{\kappa}{2} i\psi \dot{x}^m) \epsilon e^{-1}$
$\delta e = i\kappa \epsilon \psi$	$\delta \psi = \frac{2}{\kappa} \dot{\epsilon}$

→ Inf. under rep. symm.

$$\delta x^m = \kappa \dot{x}^m, \quad \delta \dot{x}^m = \kappa \ddot{x}^m, \quad \delta e = \frac{d}{d\tau}(\kappa e)$$

$$\delta \psi = \frac{d}{d\tau}(\kappa \psi), \quad \kappa = \kappa(\tau)$$

Proof: (Noether Method)

Start w/ $S^F = \frac{1}{2} \int d\tau (\dot{x}^m \dot{x}^m - i \dot{x}^m \dot{x}^m) \eta_{m,n}$

$$\delta x^m = i\epsilon \dot{x}^m \quad | \quad \delta \dot{x}^m = \dot{x}^m \epsilon$$

- We introduce the supergravity fields (h, ψ)

which have the susy trans.

$$\delta h = i\epsilon \psi \quad | \quad \delta \psi = 0$$

$$S_{\text{susy}} = \frac{1}{2\kappa^2} \int d\tau (\kappa R - \psi_m \gamma^{mnp} D_n \psi_p)$$

with susy

$$\delta h = i\epsilon \psi, \quad \delta \psi = 0$$

- The fields also have a local abelian trans.

$$\delta h = \dot{g} \quad | \quad \delta \psi = \dot{\eta}$$

$$g_i \in C^\infty(M), \quad \eta_i \in C^\infty(M)$$

* We now let ϵ become local: $\epsilon = \epsilon(\tau)$

S^F is no longer susy invariant.

$\delta_{\text{susy}} S^F \neq 0$. Instead, we find

$$S^1 \equiv S^F - i \int d\tau \frac{\kappa}{2} \psi \dot{\psi}$$

Is inv. at order κ^0 , $\eta = \frac{2}{\kappa} \epsilon$

→ link local invariance to local susy

Thus,

$$S^2 = S^F - i \int dt \frac{\kappa}{2} \psi_0$$

Note the linearized gravity

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu}$$

But $g_{\tau\tau} = -e^2$

$$\therefore -e^2 = -1 + \kappa h$$

$$e^2 = 1 - \kappa h$$

$$2e \delta e = -\kappa \delta h$$

$\Rightarrow$ Come Back later... Not sure about how we're introducing a linearized grav. term.

The Coupled action in the first order formalism:

$$\int dt \left(\dot{x}^\mu p_\mu - \frac{1}{2} e p^\mu p_\mu - \frac{i}{2} \chi_\mu \dot{\chi}^\mu - \frac{i\kappa}{2} \psi \chi^\mu p_\mu \right)$$

Susy:

$$\delta x^\mu = i\epsilon \chi^\mu$$

$$\delta \chi^\mu = p^\mu \epsilon$$

$$\delta p^\mu = 0$$

$$\delta e = i\kappa \epsilon \psi$$

$$\delta \psi = \frac{2}{\kappa} \epsilon$$

The E.O.M are given by

p^μ : $\frac{\partial}{\partial t} \frac{\partial \mathcal{L}}{\partial p^\mu} - \frac{\partial \mathcal{L}}{\partial p_\mu} = 0$

$$\dot{x}^\mu - e p^\mu - \frac{i\kappa}{2} \psi \chi^\mu = 0$$

χ^μ :

$$\dot{p}^\mu = 0$$

e :

$$p^\mu p_\mu = 0$$

Both are Constraints

$\equiv \phi$

$$\psi: \boxed{\chi^\mu p_\mu = 0} \equiv \psi$$

$$\chi^\mu: -\frac{i}{2} \dot{\chi}^\mu - \frac{i}{2} \dot{\chi}^\mu + \frac{iK}{2} \psi p^\mu = 0$$

$$2\dot{\chi}^\mu + K\psi p^\mu = 0$$

$$\boxed{\dot{\chi}^\mu - \frac{K}{2} p^\mu \psi = 0}$$

* If we eliminate p^μ via its E.O.M then we recover the original action/trast's

Gauge choices: $e=1, \psi=0$

$$\dot{p}^\mu = 0, \dot{\chi}^\mu = 0, p^\mu = \dot{x}^\mu$$

Hamiltonian approach:

$$\{x^\mu, p^\mu\} = \eta^{\mu\nu} \quad \& \quad \{\chi^\mu, \chi^\nu\} = i\eta^{\mu\nu}$$

$$H = \frac{1}{2} e p_\mu p^\mu + i \frac{K}{2} \psi p_\mu \chi^\mu = a\phi + b\psi$$

$$\frac{d\phi}{d\tau} = \{\phi, H\}$$

where:

$$\{\phi, \psi\}_{\text{pass}} = 0 \quad \& \quad \{\phi, \psi\}_p = i\phi$$

Dirac Method: Momentum conj. to χ^μ is $\frac{i}{2} \chi^\mu$
with $p_\chi^\mu = \frac{i}{2} \chi^\mu$ we must adopt the constraint

$$\boxed{S^\mu \equiv p_\chi^\mu - \left(\frac{i}{2}\right) \chi^\mu = 0}$$

Starting from

$$\{\chi^\mu, p_\chi^\nu\} = \eta^{\mu\nu}$$

we discover $S^\mu = 0$ is second class so the

$\{\phi, S^\mu\} \neq 0$ (another constraint.)

* We instead use the Dirac Bracket

$$[x^m, p^n] = i\eta^{mn}, \quad \{x^m, x^n\} = -i\eta^{mn}$$

$$x^m \rightarrow \hat{x}^m, \quad \underline{t=1}$$

$$p_m \rightarrow -i \partial / \partial x^m$$

where we may adopt the following

$$\boxed{x^m = i \gamma^m / \sqrt{2}} \quad \{\gamma^m, \gamma^n\} = 2\eta^{mn}$$

Constraint S: $\phi = p_m p^m = \frac{\partial}{\partial x^m} \frac{\partial}{\partial x_m} = -\partial^2$

$$\phi = p_m x^m = \left(-i \frac{\partial}{\partial x^m}\right) \left(i \gamma^m / \sqrt{2}\right) \quad \boxed{\phi = -\partial^2}$$

$$= \partial / \sqrt{2}$$

$$\therefore \boxed{\phi = \partial / \sqrt{2}}$$

* Impose: $\partial \gamma = 0$ (γ : wave fⁿ)

→ This is a massless spin-(1/2) particle

* $N=1$ worldline susy describes a spin 1/2 particle upon quantization.

BRST Quant: Start from

$$\boxed{S = \frac{1}{2} \int d\tau (e^{-1} \dot{x}^\mu \dot{x}^\mu - i \dot{x}^\mu \chi^\mu - \kappa e^{-1} i \psi \dot{x}^\mu \chi^\mu) \eta_{\mu\nu}}$$

* For ~~EACH~~ local symmetry a ghost is introduced

• c and γ for reparametrisations and local susy respectively.

γ : Grassmannian even since ϵ is odd

$$\epsilon \rightarrow \lambda \gamma$$

$$y^* = -y$$

• For rep: $\tau' = \tau - K(\tau)$

$$K(\tau) \rightarrow \Delta C$$

Thus, we write out the full variations:

$$\delta x^m = i\epsilon \dot{x}^m + K \dot{x}^m = i\Delta\gamma \dot{x}^m + \Delta C \dot{x}^m$$

$$\delta x^m = \dots$$

for which:

$$\begin{aligned} \delta_B x^m &= i\Delta\gamma \dot{x}^m + \Delta C \dot{x}^m & \delta_B x^m &= \left(\dot{x}^m - \frac{iK}{2} \dot{x}^m \right) \Delta\gamma \dot{x}^m \\ \delta_B e &= iK \Delta\gamma \psi + \frac{d}{d\tau}(e\Delta C) & & + \Delta C \dot{x}^m \\ \delta_B \psi &= \frac{2}{K} \frac{d}{d\tau}(\Delta\gamma) + \frac{d}{d\tau}(\Delta C \psi) \end{aligned}$$

* C : Reparametrizations
 γ : Super symmetry

• Ghost transformations:

$$\begin{aligned} \delta_C c &= \Delta C c - i\Delta\gamma^2 e^{-1} \\ \delta_B \gamma &= \frac{iK}{2} \Delta\gamma^2 \psi e^{-1} + \Delta C \gamma \end{aligned}$$

* Idea: Take the full symmetry of the theory and combine into a BRST symm.

→ Introduce Anti-ghost fields and B

b : Grassmann odd

B : Grassmann even

$$\delta b = \Delta \lambda$$

$$\delta \beta = \Delta \rho$$

- Introduce the Lagrange multipliers for each of the variables

$$b \rightarrow \lambda$$

$$\beta \rightarrow \rho$$

$$\delta \lambda = 0$$

$$\delta \rho = 0$$

- Gauge fixing terms: $\int d\tau \lambda h(e)$
since $e=1$ as well as
 $\frac{\kappa}{2} \int d\tau \psi \rho$ since $\psi=0$

$$\therefore S^{\text{gf}} = \int d\tau \left(\lambda h(e) + \frac{\kappa}{2} \rho \psi \right)$$

$$e=1, \quad \psi=0$$

- We must then add the anti-ghost field Action:

$$(i) \text{ Take } \lambda \rightarrow b, \quad \rho \rightarrow \beta \text{ in } S^{\text{gf}}$$

$$(ii) \text{ Take } S^{\text{gf}} \text{ and add } S_{\beta} S^{\text{gf}}$$

$$\int d\tau \delta_{\Delta} \left(b h(e) - \frac{\kappa}{2} \beta \psi \right) \\ = \Delta (S^{\text{gf}} + S^{\text{gh}})$$

$$S^{\text{BRST}} = S^{\text{orig}} + S^{\text{gh}} + S^{\text{gf}}$$

eliminating λ & ρ enforces the gauge cond's.

$$S^{\text{tot}} = \int d\tau \left\{ \frac{1}{2} (\dot{x}^{\mu} \dot{x}^{\mu} - i \chi^{\mu} \dot{\chi}^{\mu}) \eta_{\mu\nu} - b \dot{c} - \beta \dot{\phi} \right\}$$

$$\{c, b\} = 1 \quad \{\chi, \beta\} = 1$$

$$Q_B = c p_\mu p^\mu + \gamma p_\mu \chi^\mu + i \frac{b}{2} \gamma^2 \quad [\text{charge}]$$

BRST charge is nilpotent

$$Q^2 = 0$$

Which is the BRST generator:

$$\delta \bullet = \{ \bullet, \Delta Q \}$$

Dirac Quant: $\{c, b\} = i\hbar, \{\chi, \beta\} = i\hbar$

$$b = i \partial / \partial c, \quad \beta = -i \partial / \partial \chi$$

• Quantum BRST charge:

$$Q = -c \partial^2 - (i/2) \gamma \partial - \frac{1}{2} \gamma^2 \frac{\partial}{\partial c}$$

* Assume $\Phi \equiv \psi + \gamma \phi - \sqrt{2} i \partial \phi c$

$$Q \Phi = -i \frac{1}{2} \partial \psi \gamma - \partial^2 \psi c$$

and so we adopt

$$Q \Phi = 0 \quad [\text{Physical states}]$$

The Brink-Schwarz Superparticle:

* Space-time supersym.

$$(x^\mu, \Theta^A), \quad \mu = 0, \dots, D-1$$

Θ^A : Spacetime Majorana spinor

$$\bar{\Theta}^A \equiv \Theta^T C, \quad A = 1, \dots, 2^{D/2} \leftarrow \text{dim of cliff. alg.}$$

$$\delta x^\mu = i \bar{\epsilon} \gamma^\mu \Theta, \quad \delta \Theta = \epsilon$$

$$\partial \theta_A = \epsilon_A \quad \partial \theta_A = \epsilon_A$$

ϵ_A : constant spacetime Maj. spinor
 (x^m, θ^A) : superspace coordinates

$$[\delta_1, \delta_2]_{\text{Susy}} = (\text{spacetime translations})$$

* For two Maj. spinors: $\bar{\lambda} \gamma^m \rho = -\bar{\rho} \gamma^m \lambda$

$$\boxed{\Pi^m \equiv \dot{x}^m - i \bar{\theta} \gamma^m \dot{\theta}} \quad (\text{Susy invariant})$$

proof:

$$\begin{aligned} \delta_{\text{Susy}} \Pi &= (\delta \dot{x}^m) - i \delta \bar{\theta} \gamma^m \dot{\theta} - i \bar{\theta} \gamma^m \delta \dot{\theta} \\ &= i \bar{\epsilon} \gamma^m \dot{\theta} - i \bar{\epsilon} \gamma^m \dot{\theta} - i \bar{\theta} \gamma^m \epsilon \\ &= 0 \end{aligned}$$

* Under a Lorentz trans. of $\text{Spin}(1, D-1)$
 $\delta_L x^m = \omega_{\nu}^m x^{\nu}, \quad \delta_L \theta = \frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \theta$

Lorentz:

$$\begin{aligned} \therefore \delta_L \Pi^m &= \omega_{\nu}^m \dot{x}^{\nu} - i \left(\frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \theta \right)^T C \gamma^m \dot{\theta} \\ &= i \bar{\theta} \gamma^m \left(\frac{1}{4} \omega_{\mu\nu} \gamma^{\mu\nu} \theta \right) \\ &= \omega_{\nu}^m \dot{x}^{\nu} - i \frac{1}{4} \omega_{\mu\nu} \theta^T (\gamma^{\mu\nu})^T C \gamma^m \dot{\theta} \\ &\quad - i \frac{1}{4} \omega_{\mu\nu} \bar{\theta} \gamma^{\mu} \gamma^{\nu} \dot{\theta} \\ &= \omega_{\nu}^m \dot{x}^{\nu} \end{aligned}$$

So Π^m transforms as a vector

* We can therefore realise the following

Superspace replacement:

$$\dot{x}^m \rightarrow \Pi^m$$

$$\therefore S = \frac{i}{2} \int d\sigma e^{-1} \pi^\mu \pi^\nu \eta_{\mu\nu}$$

e is inert under susy

$$\text{E.O.M.: } (x^\mu, \theta, e)$$

$\underline{x^\mu}$:

$$\delta S = \frac{i}{2} \int d\tau e^{-1} (\dot{x}^\mu - \bar{\theta} \gamma^\mu \dot{\theta}) (\dot{x}^\mu - \bar{\theta} \gamma^\mu \dot{\theta}) \eta_{\mu\nu}$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0$$

$$\partial_\tau [e^{-1} \pi^\mu] = 0 \quad (x^\mu \text{ E.O.M.})$$

$$\underline{\theta}: \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\mathcal{L}(e^{-1} \bar{\theta} \gamma^\mu \pi_\mu) + e^{-1} \gamma^\mu \dot{\theta} \pi_\mu = 0$$

• First term vanishes since

$$= \partial_\tau (e^{-1} \pi^\mu) \bar{\theta} \gamma^\mu + e^{-1} \partial_\tau (\bar{\theta} \gamma^\mu) \pi_\mu$$

$$= \partial_\tau (\bar{\theta} \gamma^\mu) e^{-1} \pi^\mu = \dot{\theta}^\tau \gamma^\mu e^{-1} \pi^\mu$$

$$\therefore \gamma^\mu \pi_\mu \bar{\theta} = 0 = \not{\pi} \bar{\theta} \quad (\theta \text{ E.O.M.})$$

$$\underline{e}: -\frac{1}{2} e^{-2} \pi^\mu \pi^\nu \eta_{\mu\nu} = 0$$

$$\therefore \pi^\mu \pi^\nu \eta_{\mu\nu} = 0$$

Thus: E.O.M.:

$$\frac{d}{d\tau} (e^{-1} \pi^\mu) = 0, \quad \not{\pi} \bar{\theta} = 0$$

$$\pi^\mu \pi^\nu \eta_{\mu\nu} = 0$$

$\Rightarrow$ We note that $\delta \bar{\theta} \gamma^\mu \dot{\theta} = -\dot{\bar{\theta}} \gamma^\mu \delta \theta$

& The action is invariant under susy

* The corresponding first order action:

$$S = \int d\tau \left\{ \pi^\mu \dot{p}_\mu - \frac{1}{2} e \dot{p}^\mu \dot{p}_\mu \right\} \mathcal{H}_{\text{can}} \quad \textcircled{\text{A}}$$

w/ susy: $\delta e = \delta p^\mu = 0$

* π is not invertible since

$$\pi^\mu \pi_\mu = \pi^\mu \pi_\mu = 0$$

Thus we cannot conclude $\dot{\theta} = 0$

• $\textcircled{\text{A}}$ possesses an extra symmetry:

* K-symmetry:

$$\begin{aligned} \delta x^\mu &= i \bar{\theta} \gamma^\mu \dot{\theta} K, & \delta \theta &= i \dot{\theta} K \\ \delta e &= -4 e \dot{\theta} K \end{aligned}$$

K_A : is a Majorana spinor. $K_A = K_A(t)$

Claim: $\delta_K S = 0$

Proof: we first show $\delta \pi^\mu = -2 \dot{\theta} \gamma^\mu K$ (i)
and $\delta(\pi^\mu \pi_\mu) = -4 \pi^\mu \pi_\mu \dot{\theta} K$ (ii)

$$\begin{aligned} \text{(i)} \quad \delta \pi^\mu &= \delta \dot{x}^\mu - i \delta \bar{\theta} \gamma^\mu \dot{\theta} - i \bar{\theta} \gamma^\mu \delta \dot{\theta} \\ &= \frac{d}{d\tau} (i \bar{\theta} \gamma^\mu \dot{\theta}) - i \delta \bar{\theta} \gamma^\mu \dot{\theta} - i \bar{\theta} \gamma^\mu \delta \dot{\theta} \end{aligned}$$

(ii) $\rightarrow$ trivial

• We note that the first order formalism

$$\int (\pi^\mu \dot{p}_\mu - \frac{1}{2} e \dot{p}^\mu \dot{p}_\mu) d\tau$$

is also K-inv. if we also impose

δx^μ and e : E.O.M but take

$$\delta \theta = i \dot{\theta} K, \quad \delta p^\mu = 0$$

* The K -symm forms a closed alg. if we are on-shell ~~and~~ impose
 $\delta X^m = i \bar{\theta} \gamma^m \delta \theta$, $\delta \theta = L(\tau) \dot{\theta}$, $\delta e = 0$

($\Rightarrow$) There is NO Lorentz covariant quant. for the Brink-Schwarz particle

* Conjugate Momenta of θ^A : λ^A

$$\lambda^A \equiv \frac{\partial \mathcal{L}}{\partial \dot{\theta}^A}$$

• first note that $P_\mu = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu}$
 $\therefore$ $e^{-1} T_{\mu} = P_\mu$

$$\begin{aligned} \delta \mathcal{L} / \delta \dot{\theta}^A &= i (\bar{\theta} \gamma^\mu \delta \theta^A) P_\mu \\ &= i (\bar{\theta} \gamma^\mu)^A P_\mu \quad \text{or} \\ &= i (\bar{\theta} \not{p})^A \end{aligned}$$

$\therefore$

$\lambda^A = i (\bar{\theta} \not{p})^A$

Since $\frac{\partial}{\partial \bar{\theta}} (\bar{\theta} \gamma^\mu \theta) = - \bar{\theta} \gamma^\mu \frac{\partial \theta}{\partial \bar{\theta}}$

* This tells us that λ^A and θ^A are related, thus we can view this as an extra constant of the system constant

$\bar{\chi}^A \equiv \bar{\lambda}^A - i (\bar{\theta} \not{p})^A = 0$

• Factor of i causes problems as it is not

invertible

$$\{\theta_A, \bar{\lambda}^B\}_P = \delta_A^B \quad | \quad \{\bar{\chi}^A, \bar{\chi}^B\} = -2i(\not{C}\not{P})^{AB}$$

$$\{\chi_A, \bar{\chi}^B\} = -2i(\not{P})_A^B$$

Super Space formulation:

(τ, θ) : Superspace of pt particle w/ worldline
susp.

- Supercharges were given by

$$\vec{j} = \vec{Q} = \dot{x}^\mu \chi^\nu \eta_{\mu\nu}$$

In Particular, $[\delta_1, \delta_2] = [\epsilon_1 Q_1, \epsilon_2 Q]$ closes to
a time translation.

$$[\delta_1, \delta_2] = -\epsilon_1 \epsilon_2 \{Q, Q\} = 2i \epsilon_2 \epsilon_1 P$$

$$P \equiv \frac{d}{d\tau}$$

Thus, we have the following supersym alg.

$$\{Q, Q\} = +2iP, \quad [Q, P] = 0$$

Since Q and P are Generators we can consider
the supergroup generated by these:

$$g(\theta, \tau) = \exp(\theta Q + \tau P)$$

* Acting w/ $g(a, \epsilon) = e^{-\epsilon Q - aP}$, the pt
 $(\tau, \theta) \mapsto (\tau', \theta')$

$$g(a, \epsilon)g(\theta, \tau) = (1 - \epsilon Q - aP)(1 + \theta Q + \tau P)$$

$$= (1 - \epsilon Q - aP + \theta Q + \tau P - \epsilon Q \theta Q - a\tau P^2$$

$$- aP \theta Q - \epsilon Q \tau P)$$

$$= 1 + (\tau - a)P + (\theta - \epsilon)Q - a\tau P^2 - aP\theta Q - \epsilon Q\tau P$$

OK to first order:

$$e^x e^y = e^{x+Y + \frac{1}{2}[x,Y] + \mathcal{O}(x^2, y^2)}$$

$$\textcircled{a} = [-\epsilon Q - aP, \theta Q + \tau P] \\ = -[\epsilon Q, \theta Q]$$

all other commutators vanish.

$$\begin{aligned} -[\epsilon^\alpha Q_\alpha, \theta^\beta Q_\beta] &= -(\epsilon^\alpha Q_\alpha \theta^\beta Q_\beta - \theta^\beta Q_\beta \epsilon^\alpha Q_\alpha) \\ &= -\epsilon^\alpha \theta^\beta Q_\alpha Q_\beta - \epsilon^\alpha \theta^\beta Q_\beta Q_\alpha \\ &= \epsilon^\alpha \theta^\beta \{Q_\alpha, Q_\beta\} \end{aligned}$$

$$\textcircled{a} = \frac{i}{2} \epsilon \theta \{Q, Q\} = i \epsilon \theta P$$

$\therefore$

$$g(\tau, \epsilon) g(\tau, \theta) = g(\tau - a + i \epsilon \theta, \theta - \epsilon)$$

Which means that such a group action will move us along SuperSpace:

$$\tau' = \tau - a + i \epsilon \theta$$

$$\theta' = \theta - \epsilon$$

* Def: A Superfield is a field which is dependent on (τ, θ) . The field has a "Scalar" nature to it. Reg'd

$$\Phi'(\tau', \theta') = \Phi(\tau, \theta)$$

It is easy to generalize, as with $\mathbb{R}$ -scalar fields

$$\delta \Phi = \Phi'(\tau, \theta) - \Phi(\tau, \theta) = (aP + \epsilon Q) \Phi$$

$$\begin{aligned}
g(\bar{u} \in) \Phi &= \Phi - \delta \Phi \\
\Phi'(\tau', \theta') &= \Phi'(\tau', \theta) - \epsilon \frac{d}{d\theta} \Phi'(\tau', \theta) \\
&= \Phi'(\tau, \theta) + (-a + i\epsilon\theta) \frac{d}{dt} \Phi(\tau, \theta) - \epsilon \partial_\theta \\
&= \Phi'(\tau, \theta) + (-a + i\epsilon\theta) \partial_t \Phi \\
&\quad - \epsilon \frac{d}{d\theta} \left[\Phi'(\tau, \theta) + (-a + i\epsilon\theta) \frac{d}{dt} \Phi(\tau, \theta) \right] \\
&= \Phi'(\tau, \theta) + (-a + i\epsilon\theta) \partial_t \Phi - \epsilon \partial_\theta \Phi + \cancel{\epsilon^2} \\
&= \Phi'(\tau, \theta) - a \partial_t \Phi + i\epsilon\theta \partial_t \Phi - \epsilon \partial_\theta \Phi \\
&= \Phi(\tau, \theta)
\end{aligned}$$

$$\begin{aligned}
\therefore \delta \Phi &= \Phi'(\tau, \theta) - \Phi(\tau, \theta) \\
&= a \partial_t \Phi + \epsilon \left(\partial_\theta - i \frac{d}{dt} \theta \right) \Phi
\end{aligned}$$

We conclude

$$P = \partial_t$$

$$Q = \frac{d}{d\theta} - i \frac{d}{dt} \theta$$

* We define the covariant derivative:

$$D = \partial_\theta + i d\tau \theta$$

s.t.

$$\begin{aligned}
\{D, Q\} &= 0, \quad [d\tau, Q] = 0 \\
\{D, D\} &= 2i d\tau
\end{aligned}$$

We now take a Scalar Superfield $\mathbb{X}(\bar{u}, \theta)$

$$\text{We know } f(\theta) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \theta^n$$

But since $\theta^2 = 0$

$$f(\theta) = f(0) + \theta f^{(1)}(0)$$

Now

$$\mathbb{X}(\tau, \theta) = \mathbb{X}(\tau, 0) + \left(\frac{d}{d\theta} \mathbb{X}(\tau, \theta) \right) \Big|_{\theta=0} \theta$$

Since $\mathbb{X}$ is a scalar field then

$\mathbb{X}(\tau, 0)$ must be Grassmann even

while $(\delta \theta X)$ is odd.

$$\Rightarrow X(\tau, 0) \equiv X(\tau)$$

$$\delta X(\tau, 0)|_{\theta=0} \equiv \dot{X}(\tau) i$$

Therefore:

$$X(\tau, \theta) = X(\tau) + i\theta \dot{X}(\tau)$$

* Note $(\delta \bullet = \epsilon Q \bullet = \epsilon D \bullet)|_{\theta=0}$

ex:

$$\delta X = \epsilon Q X|_{\theta=0} = \epsilon D X|_{\theta=0} = \epsilon i \dot{X}(\tau)$$

as well as:

$$\delta X = \delta(i D X|_{\theta=0}) = -i \epsilon Q D X|_{\theta=0}$$

at $\theta=0$ $Q \equiv D$

$$\begin{aligned} \delta X &= -i \epsilon D D X|_{\theta=0} = \frac{-i\epsilon}{2} \{D, D\} X|_{\theta=0} \\ &= \frac{-i}{2} i \frac{d}{d\tau} X|_{\theta=0} = \epsilon \dot{X} \end{aligned}$$

$\therefore$

$$\delta X(\tau) = i \epsilon \dot{X}(\tau)$$

$$\delta X(\tau) = \dot{X} \epsilon$$

Note $\int d\theta X = i \dot{X}(\tau)$ as well as

$$\int d\tau d\theta = \int d\tau \frac{d}{d\theta} = \int d\tau D$$

if acting on any superfield

• For any scalar superfield Φ

$\int d\tau d\theta \Phi$ is susy inv.

• The original Action S for worldline susy

is given by $(-i/2) \int d\tau d\theta \dot{X} D X$

Proof of invariance:

$$\delta \int d\tau d\theta X = \int d\tau d\theta (aP + eQ)X$$

We assume each field has compact support so that

$$\textcircled{a} \int d\tau \frac{d}{d\tau} (aX) = aX|_{\partial R} = 0$$

$$\textcircled{b} \int d\tau d\theta e (\partial_\theta - i\partial_\tau \theta) (X + i\theta X(\tau)) \\ = \underbrace{\int d\tau d\theta iX(\tau)}_{\text{No } \theta \rightarrow 0} - \underbrace{\int d\tau d\theta i\theta \frac{d}{d\tau} X(\tau)}_{\text{compact support}}$$

Proof of Expression:

$$-\frac{i}{2} \int d\tau d\theta \dot{X} D X = -\frac{i}{2} \int d\tau D(\dot{X} D X) \\ = -\frac{i}{2} \int d\tau [D \dot{X} D X + \dot{X} D^2 X]$$

$$D X = (\partial_\theta + i \frac{d}{d\tau} \theta) (X + i\theta X) \\ = i\dot{X} + i\theta \ddot{X} \quad (\text{again, } \theta^2 = 0)$$

$$D \dot{X} = i\ddot{X} + i\theta \ddot{\dot{X}}$$

$$D(DX) = (\partial_\theta + i \frac{d}{d\tau} \theta) (i\dot{X} + i\theta \ddot{X}) \\ = i\ddot{X} - \theta \ddot{\dot{X}}$$

$\therefore$

$$D \dot{X} D X = (i\ddot{X} + i\theta \ddot{\dot{X}}) (i\dot{X} + i\theta \ddot{X}) \\ = -\ddot{X} \dot{X} - \ddot{\dot{X}} \theta \dot{X} - \theta \ddot{\dot{X}} \ddot{X}$$

$$\dot{X} D^2 X = (\dot{X} + i\theta \ddot{X}) (i\ddot{X} - \theta \ddot{\dot{X}}) \\ = i\ddot{X} \dot{X} - \theta \ddot{\dot{X}} \dot{X} - \theta \ddot{X} \ddot{\dot{X}} \quad \left. \begin{array}{l} \text{Int.} \\ \text{by} \\ \text{parts} \end{array} \right\}$$

$$-\frac{i}{2} \int d\tau (i\ddot{X} \dot{X} + \dot{X} \ddot{\dot{X}}) = \frac{1}{2} \int d\tau (\dot{X} \ddot{X} - i\dot{X} \ddot{\dot{X}}) \quad \checkmark$$

Super Gravity Superfield:

$$\mathbb{E} = e(1 + i\theta\psi)$$

Coupling the scalar field X to (super)gravity

$$(-i/2) \int d\tau d\sigma E^{-1} \bar{\psi} D\psi = S_E$$

Provided we make the rescaling

$$\psi \rightarrow e^{c/2} \psi, \quad X \rightarrow e^{c/2} X$$

$$E = \det(E_\alpha^\mu)$$

* Claim: S_E is Superconformally invariant